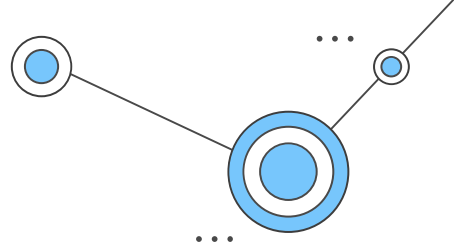


Compute Service



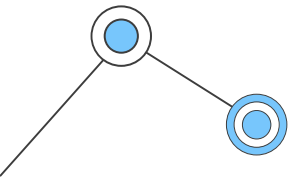
Compute Services



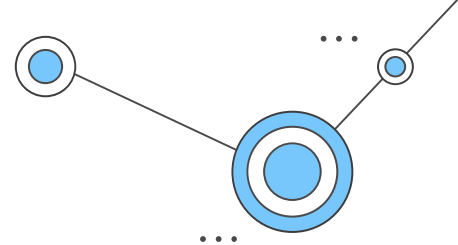
Provision of computing power on demand

Computing power to run applications/code in the cloud

- Azure VMs + Scale Sets + Availability Sets
- DevTest Labs
- Azure Virtual Desktop
- Azure Container Instances
- Azure App Service
- Azure Functions + Azure Logic Apps (or serverless computing)

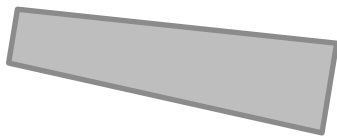
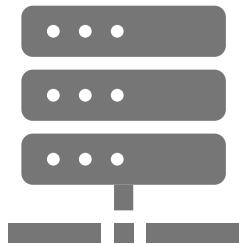


Virtual Machines (VMs)



IaaS
Infrastructure-as-a-Service

Software emulations of physical computers/servers



Virtual Processor

Virtual storage

Virtual memory

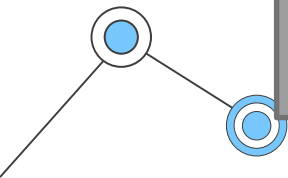


Use case:
Missing physical server

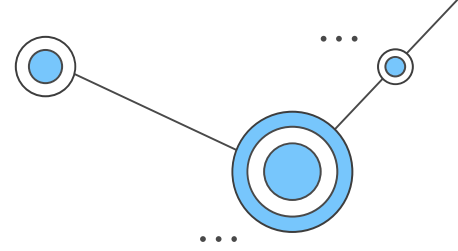


All software is fully customizable

Operating system (Windows / Linux etc.)



Virtual Machines (VMs)

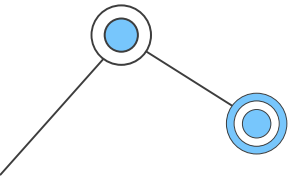


Benefits

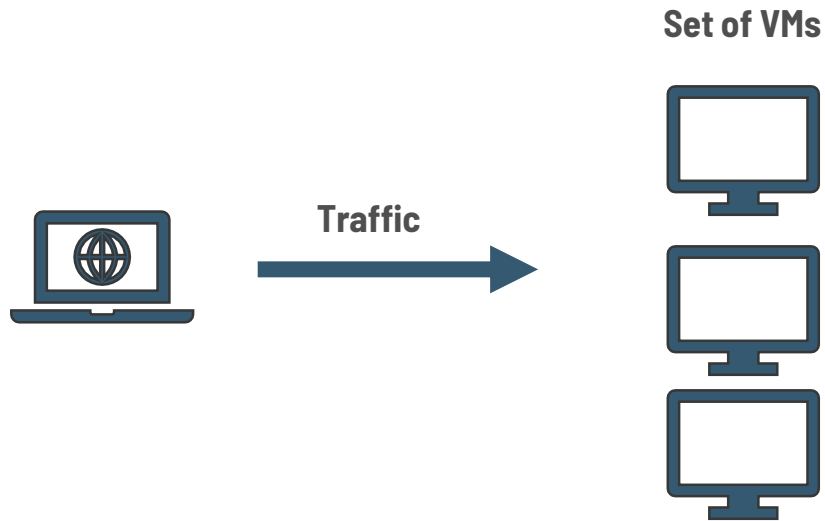
- Control of operating system
- No need of buying hardware
- Possibility to run custom applications

⇒ *It is necessary to configure, update, and maintain all software that runs on the VM*

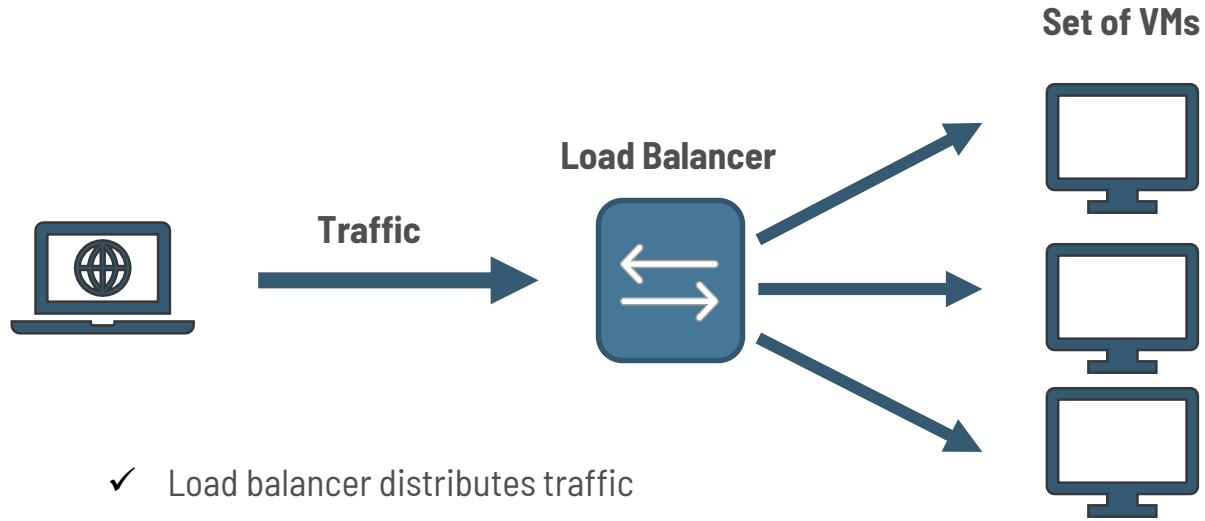
- ✓ Create and use images: Template with preconfigure OS and software



Load Balancers

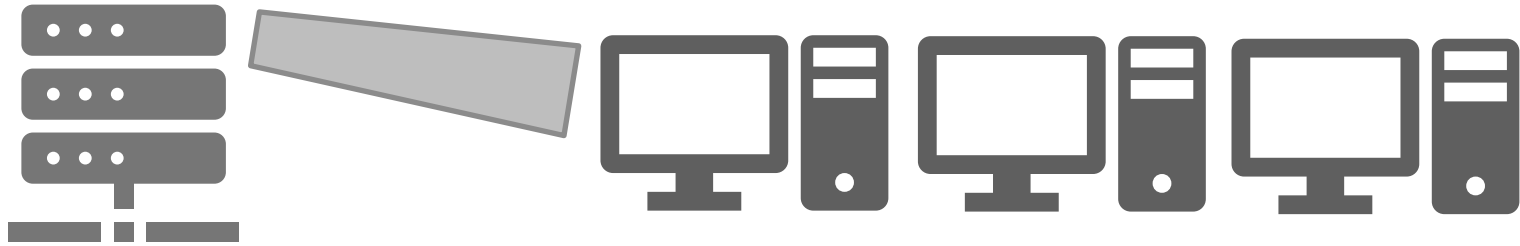
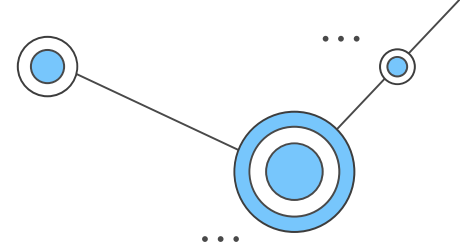


Load Balancers



- ✓ Load balancer distributes traffic
- ✓ Increase the availability and network performance

Virtual Machine Scale Sets

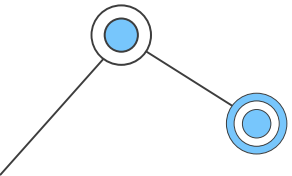


Deploy a group of identical VMs

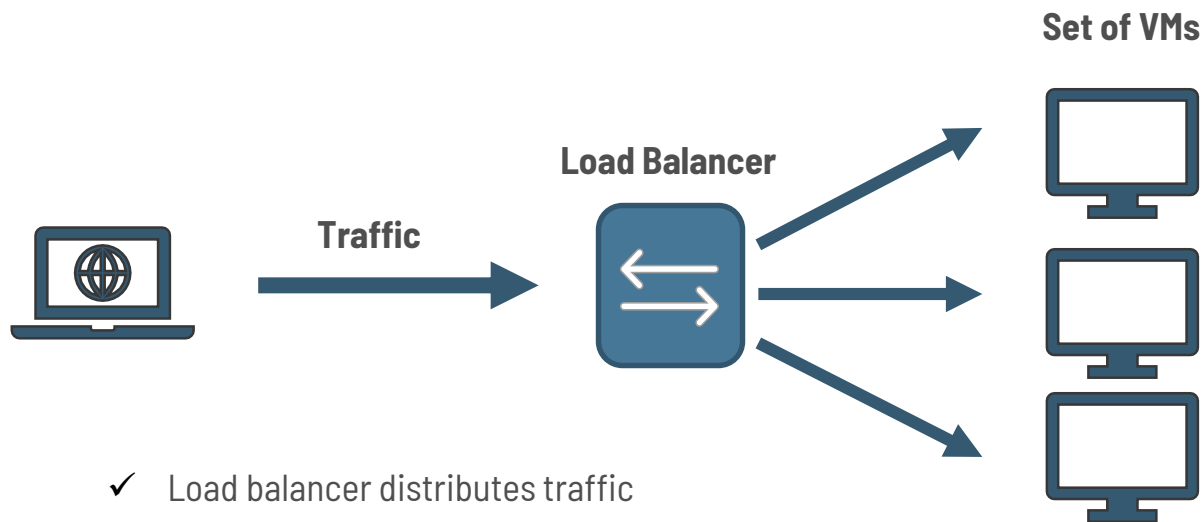
All VMs in a VM scale set are configured the same

Facilitates the creation of large resources that rely on high computing power

Manual or automatic adjustment to demand (scaling)

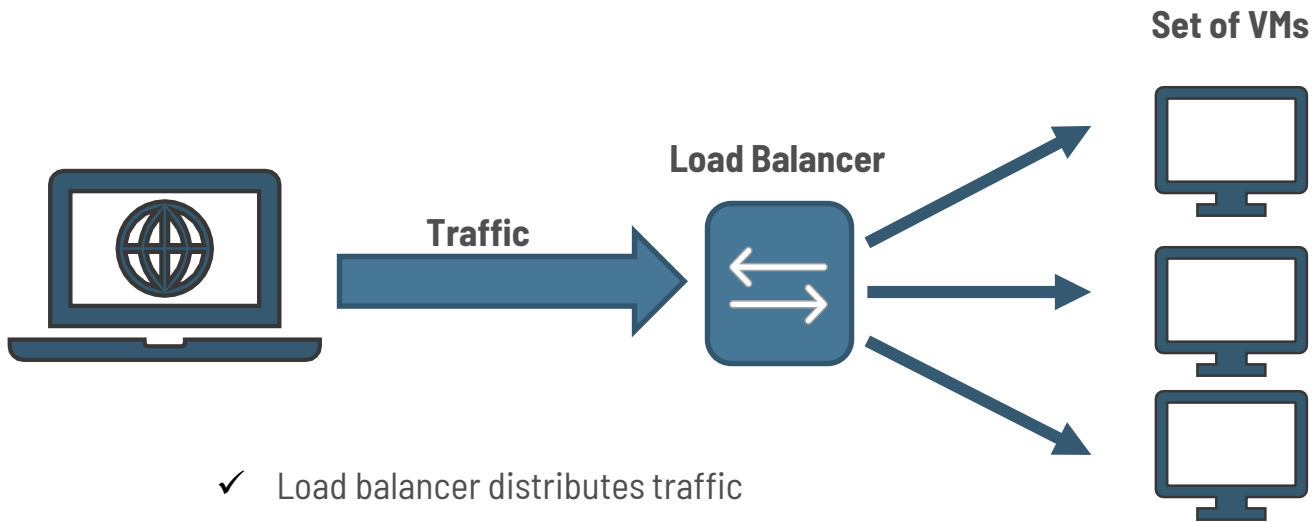


Virtual Machine Scale Sets



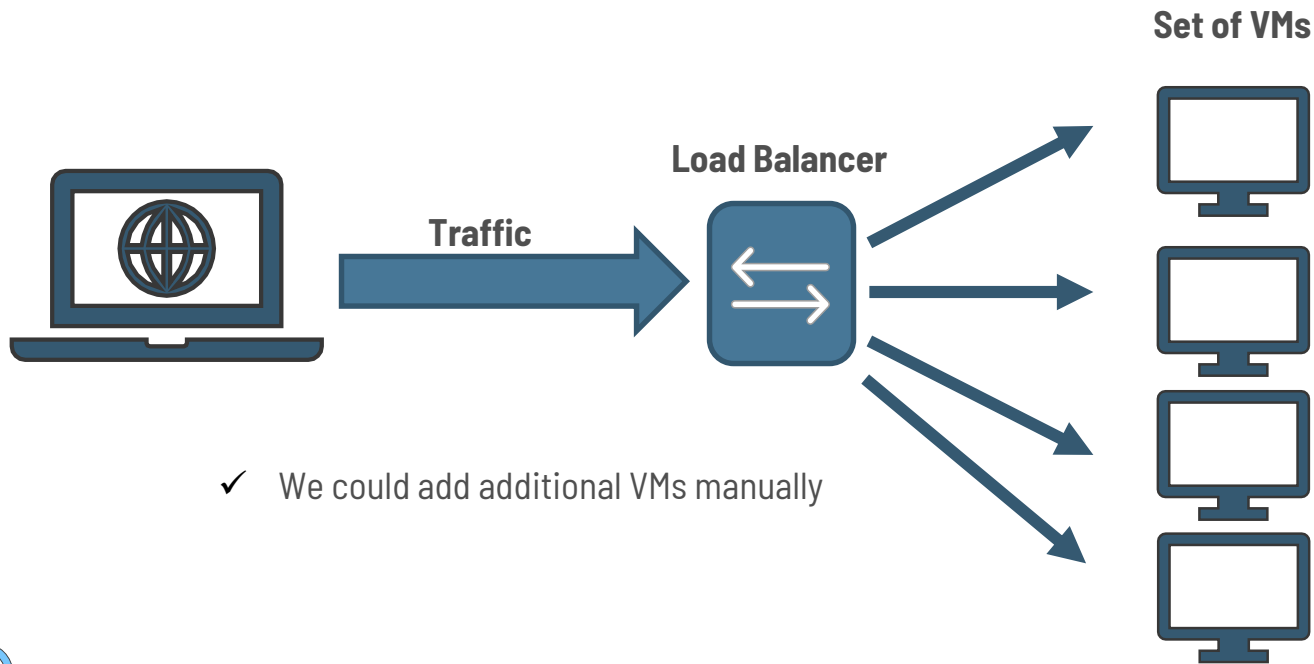
- ✓ Load balancer distributes traffic
- ✓ Increase the availability and network performance

Virtual Machine Scale Sets

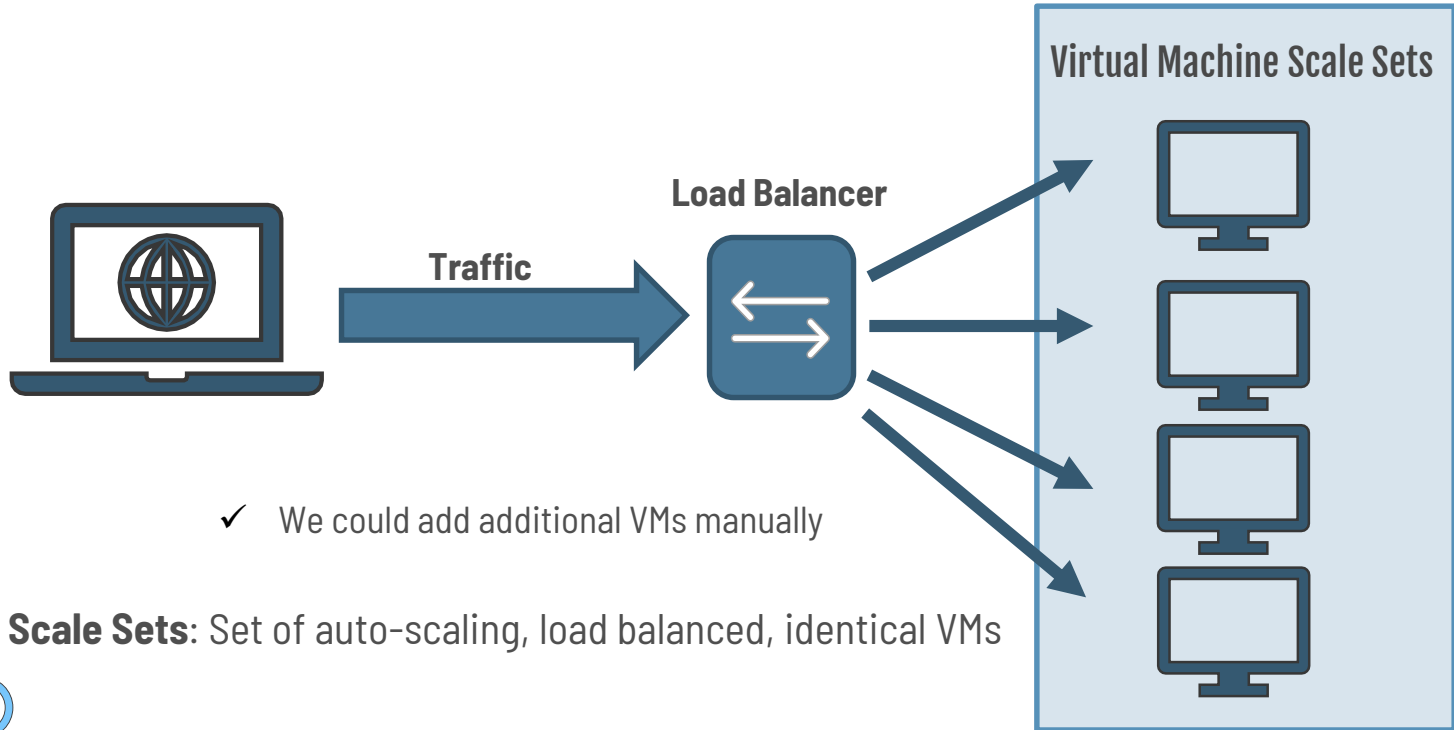


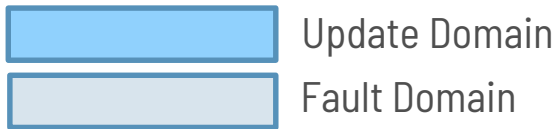
- ✓ Load balancer distributes traffic
- ✓ Increase the availability and network performance

Virtual Machine Scale Sets

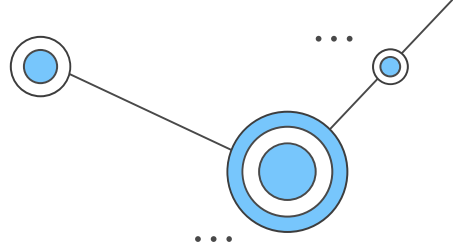


Virtual Machine Scale Sets

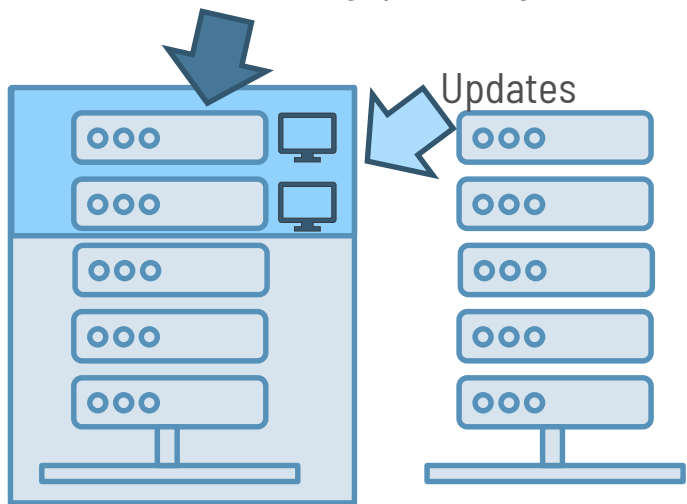




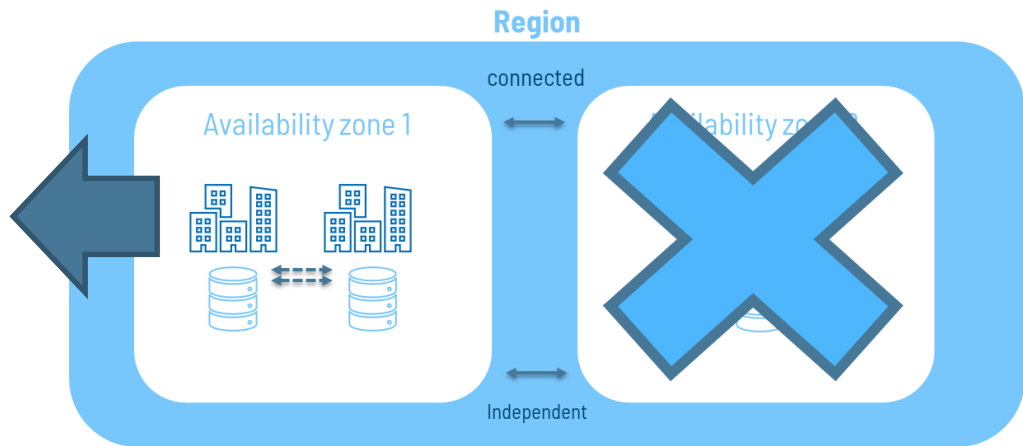
Availability Sets



Connected cooling, powering, networking

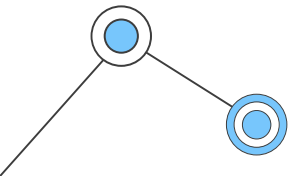


Availability Zone: Protection from entire data center failure

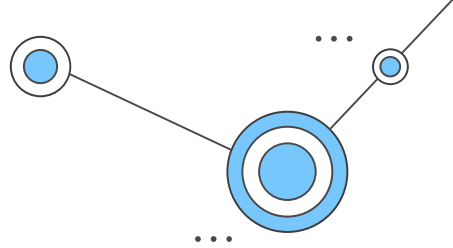


Availability sets protection against failure within data center (rack wide failure)

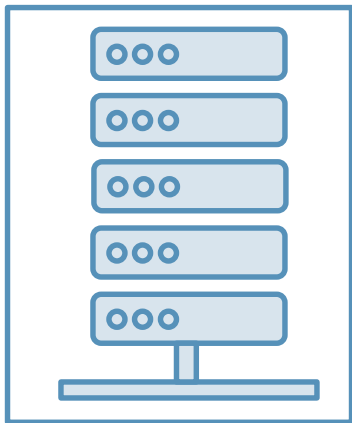
Availability sets group VMs inside a single data center



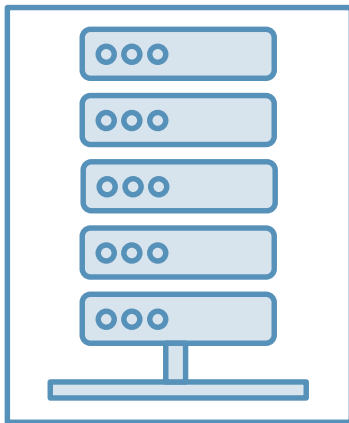
Availability Sets



Fault Domain 0



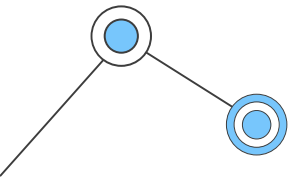
Fault Domain 1



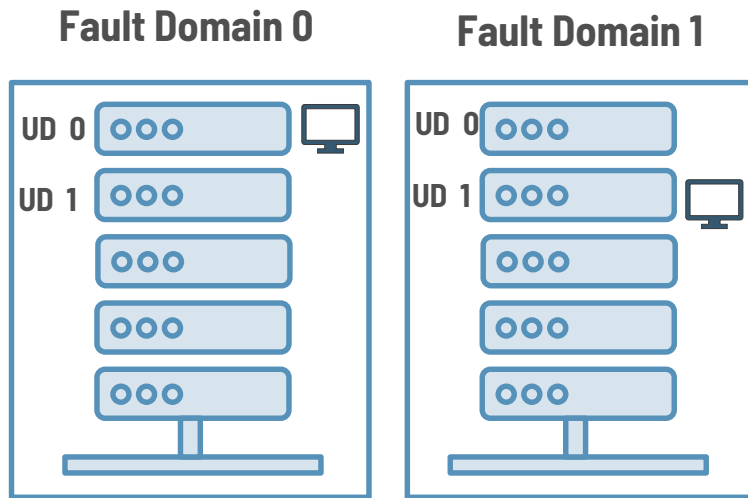
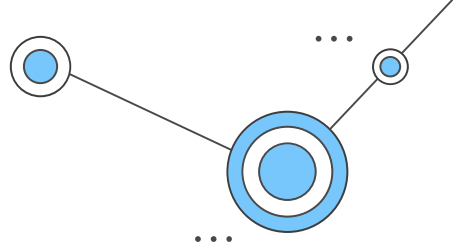
 Update Domain
Can be rebooted together

 Fault Domain
Independent cooling, powering, networking

Availability sets group VMs inside a single data center



Availability Sets

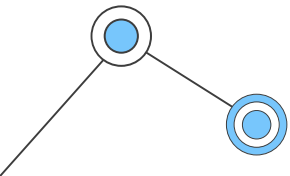


Update Domain
Can be rebooted together

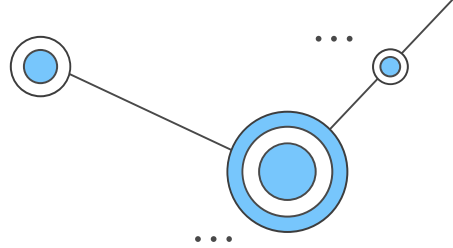
Fault Domain
Independent cooling, powering, networking

Availability sets group VMs inside a single data center

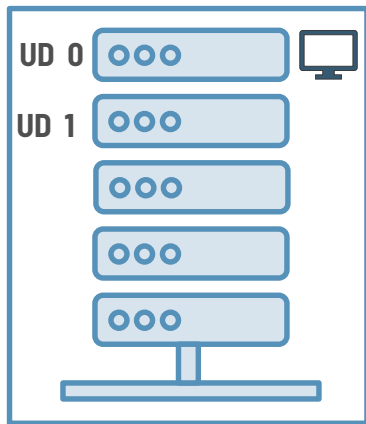
VMs will be automatically distributed across FD and UD



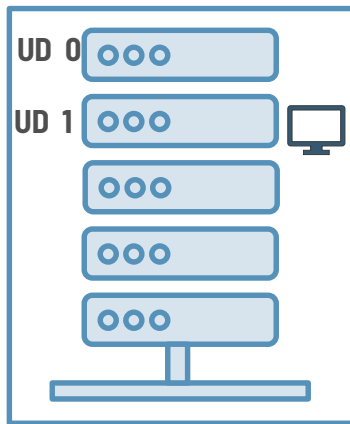
Availability Sets



Fault Domain 0



Fault Domain 1



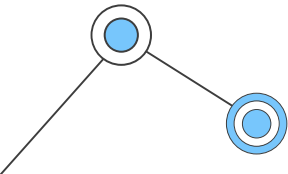
 Update Domain
Can be rebooted together

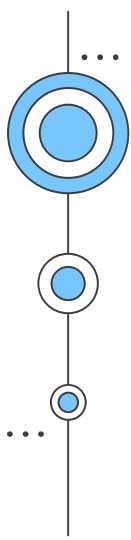
 Fault Domain
Independent cooling, powering, networking

Guaranteed availability:

2 or more VMs within Availability Sets: 99.95%

2 or more VMs across 2 Availability Zones: 99.99%



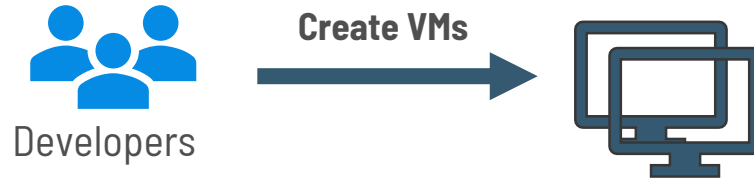


DevTest Labs



DevTest Labs

Idea: Provide easy access to creating VMs for development and testing



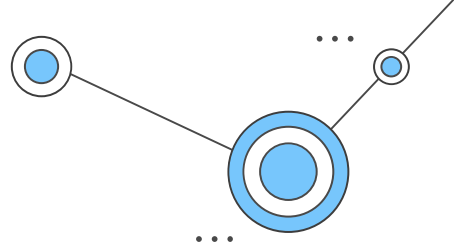
DevTest Lab users can easily & quickly create VMs

Pre-configured VMs with pre-installed development tools

To control cost: Need to follow defined policies

Use-cases: Development, testing, training

Azure Virtual Desktop



Benefits

Independent from hardware

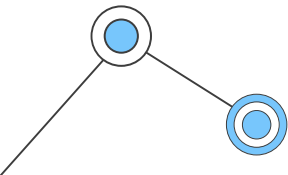
Access to application or to entire desktop

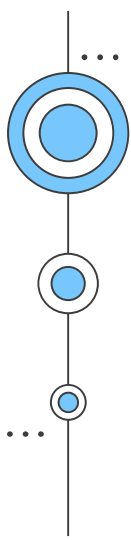
Centralized security:

Operating system, apps and data are separated from your local hardware

Risk of confidential data left on hardware is avoided

Cloud security features like MFA





Azure Virtual Desktop



Azure Virtual Desktop

Desktop and app virtualization – accessible through a browser



We can access a cloud-hosted version of Windows from

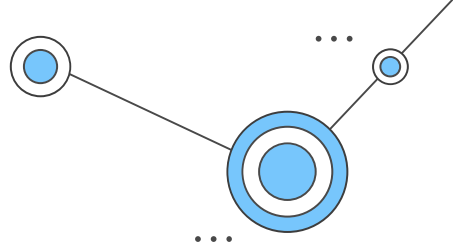
- any location
- any device and
- any operating systems

Multiple operating systems are possible:

- Windows 10,
- Windows 7,
- Windows 11,
- Windows Server

Allows multiple concurrent user-sessions

Azure Virtual Desktop



Benefits

Independent from hardware

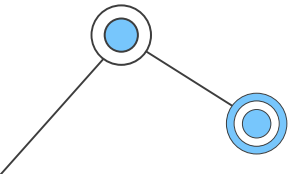
Access to application or to entire desktop

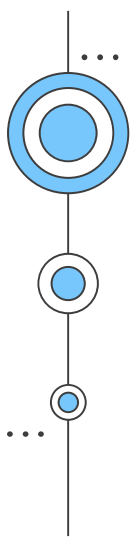
Centralized security:

Operating system, apps and data are separated from your local hardware

Risk of confidential data left on hardware is avoided

Cloud security features like MFA



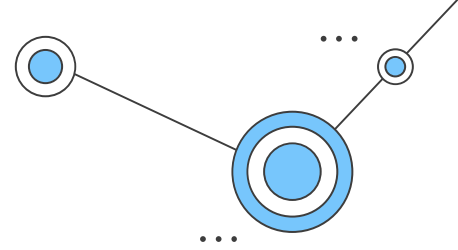


App Service



App Service

Platform-as-a-service to deploy and host web applications



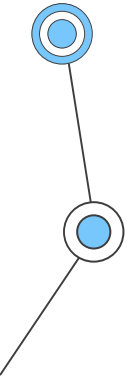
Focus on development of application without worrying about the infrastructure

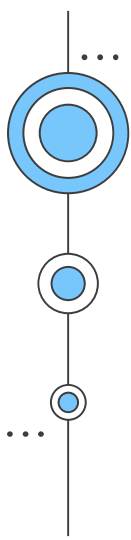
Managed security & autoscaling

Pay only for compute resources used, according to the selected app services plan (Free, different paid ones)

Programming in many languages e.g. .NET, .NET Core, Java, Ruby, Node.js, PHP or Python

Continuous deployment, e.g. via Azure DevOps





Container Services



Container Services

Application development moving towards **microservices**

One complex application / service

**Environment
can be complex
to manage**

Environment?

Python?

Java?

Operating system?

Libraries?

Registration

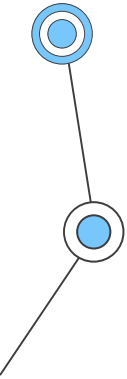
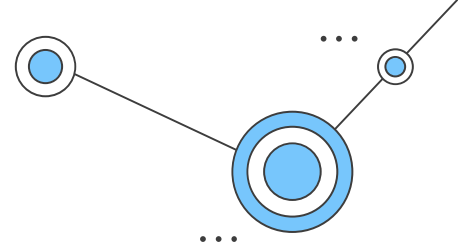
Messaging

Database service

API Gateway

**Loosely connected
microservices**

**Enables rapid delivery
of complex applications**



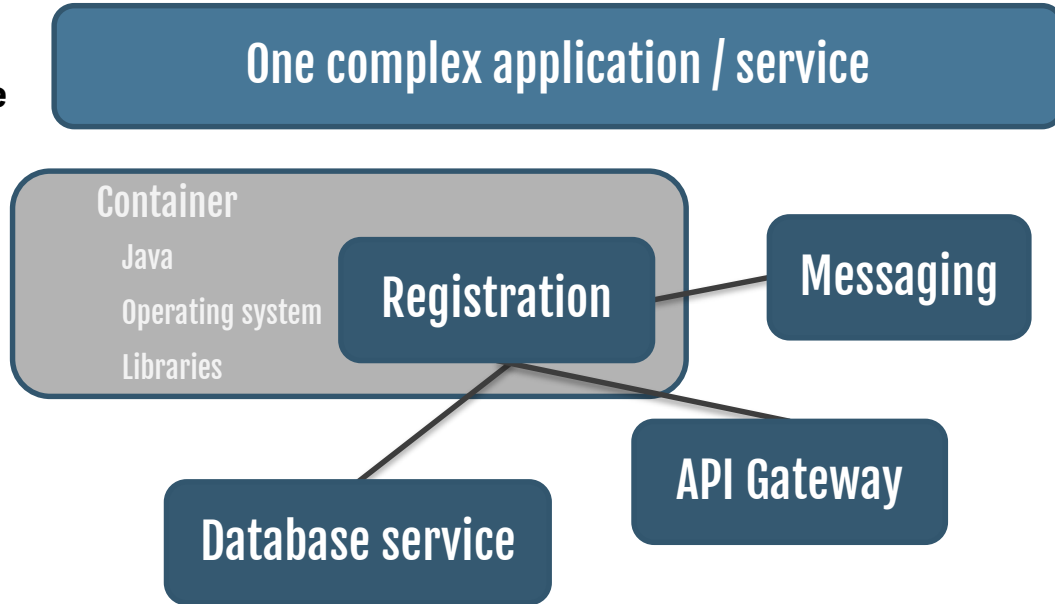
Container Services

Containers package software for deployment

Containers
package software
for deployment

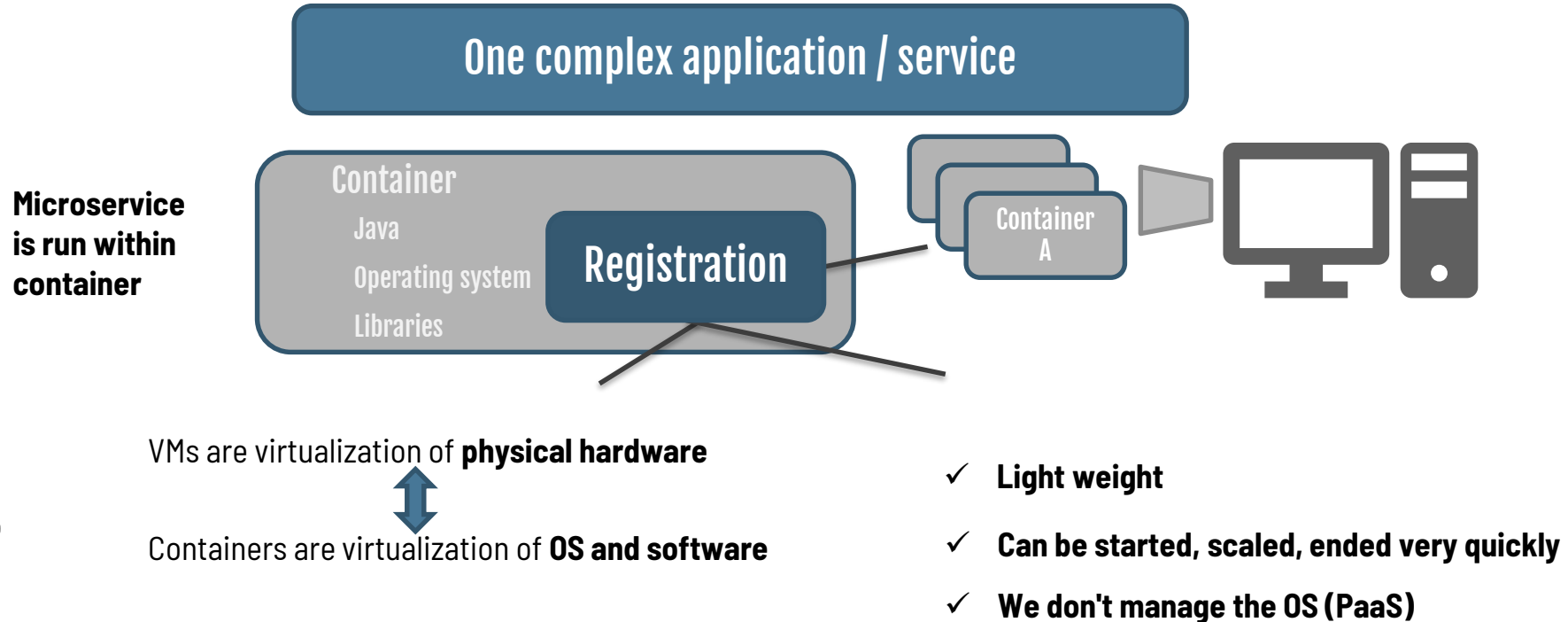


Microservice
is run within
container

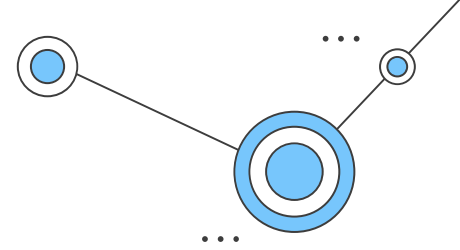


Container Services

Containers package software for deployment



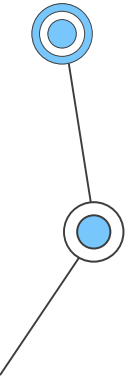
Container Services

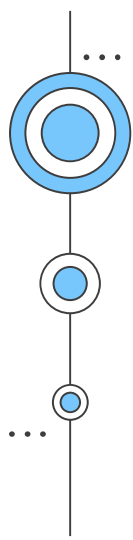


Docker: One of the most popular container engines

Azure Container Instances (ACI):

- PaaS
- Fast and simple way to upload & run containers
- No need to manage a virtual machine

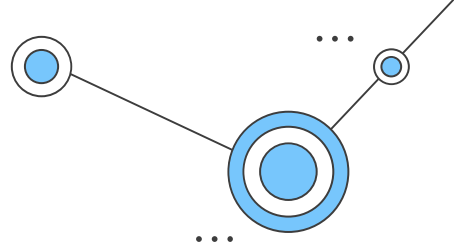




Azure Kubernetes Service



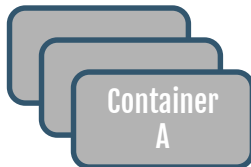
Azure Kubernetes Service



Azure Container Instances

Enables quick and easy deployment and management of containers without VMs

Containers at scale

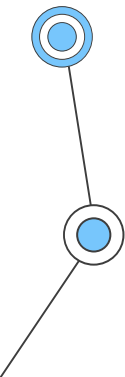


Azure Kubernetes Service

Open-source **orchestration services** to deploy, manage, and scale containers

Manage and deploy containers at scale

Quickly create and scale containers



Azure Virtual Desktop

Desktop and app virtualization – accessible through a browser



We can access a cloud-hosted version of Windows from

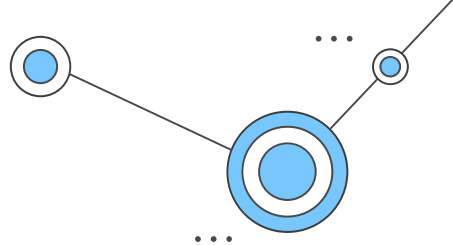
- any location
- any device and
- any operating systems

Multiple operating systems are possible:

- Windows 10,
- Windows 7,
- Windows 11,
- Windows Server

Allows multiple concurrent user-sessions

Azure Virtual Desktop



Benefits

Independent from hardware

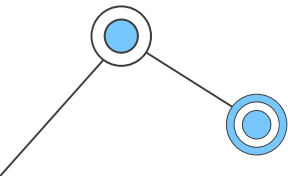
Access to application or to entire desktop

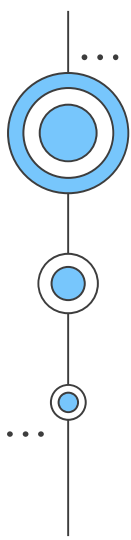
Centralized security:

Operating system, apps and data are separated from your local hardware

Risk of confidential data left on hardware is avoided

Cloud security features like MFA

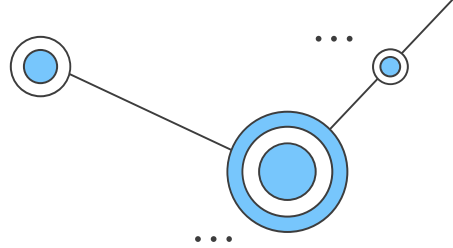




Summary



Summary



Virtual Machines

Virtualization of physical server/computer

Infrastructure-as-a-service

All software + OS is fully customizable

Fully responsible to maintain all software

DevTest Labs

Enables users to easily create pre-defined VMs for development and testing

VM Scale Sets

Set of auto-scaling, load balanced, identical VMs

Availability Sets

Group VMs inside a single data center
into Fault & Update Domains

Protection against failure within data center (rack wide failure)

Azure Virtual Desktop

Desktop and app virtualization – accessible through a browser

Virtualization of different operating systems are possible

Operating system, apps and data are separated from your local hardware

Allows multiple concurrent user-sessions



Summary

App Service

Platform-as-a-service

Deploy and host web applications

Managed security & autoscaling

Azure Container Instances

Platform-as-a-service

Containers package software for deployment

Fast and simple way to upload & run containers

No need to manage a virtual machine

Azure Kubernetes Services

Orchestration service to deploy, manage, and scale containers at scale

